

Instructions

There are **30 total points**. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ☑ ● ✗ are incorrect .

1. [1 point]: Write your full name in the box to acknowledge the instructions.

Use Cases

2. [4 points]: When using DNS queries and responses to identify traffic flows for a service, which of the following pieces of information would you need from the corresponding DNS traffic? Select all that apply.

- ☐ The date and time of each query and response ☐ The DNS names corresponding to the service
☐ The DNS query ID for each query and response ☐ The latency of the query and response

3. [2 points]: A feature that we explored in the first assignment for quality of experience estimation was the number of segments per second. What is a segment?

(Answer inside the box)

4. [2 points]: Given a sequence of packets (e.g., captured from a packet capture tool like Wireshark), what property or properties of the packets could you use to group packets into segments?

(Answer inside the box)

Data Acquisition

5. [4 points]: Which characteristics of a network traffic trace are used to group packets into flows? Select all that apply.

- ☐ Source and destination IP addresses ☐ Source and destination port numbers
☐ Protocol ☐ Packet size
☐ Time of day ☐ Packet sequence numbers

6. [2 points]: In the context of using machine learning to predict network performance, which of the following statements best describes active performance measurements? Select all that apply.

- ☐ Sending test packets into the network. ☐ Performing a download.
☐ Capturing network traffic from users. ☐ Monitoring historical network logs.

7. [2 points]: Describe one advantage of using active measurements as a way of gathering features that could be used as input to a machine learning model for predicting network performance.

(Answer inside the box)

Initials: ____

Feature Engineering

8. [2 points]: Many machine learning models are now capable of taking “raw” data as input, such as images, text, or even network traffic traces. However, in many cases, it is still useful to perform some preprocessing on the data before feeding it into the model. Provide one reason why preprocessing inputs to a machine learning model might improve model **efficiency**.

(Answer inside the box)

9. [2 points]: Provide one reason why preprocessing inputs to a machine learning model might improve model **accuracy**.

(Answer inside the box)

Machine Learning Pipelines

10. [2 points]: In the context of the bias-variance tradeoff, which of the following scenarios best describes a model with too-high variance? Select all that apply.

- ☐ Performs poorly on the training data. ☐ Performs well on the training data but poorly on the test data.
☐ Performs well on the training data and well on the test data. ☐ Shows stable performance across different datasets.

11. [5 points]: Your team is building a machine learning model to predict network performance. You have collected a dataset of network traces, and you have preprocessed the data to extract features. One of the features is normalized throughput, which is the ratio of the throughput of a network flow to the maximum throughput observed in the dataset.

When computing this value for training, do you use:

- ☐ The training set. ☐ The entire dataset.

Briefly explain your answer.

(Answer inside the box)

Feedback

12. [1 point]: Interest (1=Boring!; 10=Amazing!):

Difficulty (1=Too easy; 10=Too hard):

13. [1 point]: 1. One thing you like. 2. One suggestion for improvement:

(Answer inside the box)

Initials: ____